Assignment-Regression Algorithm

1) Identify your problem statement

Stage 1 – Machine Learning -> Input majorities are numbers
Stage 2 – Supervised Learning -> Requirements are Clear
Stage 3 - Regression -> Output are Numerical values

2) Tell basic info about the dataset (Total number of rows, columns)

Client wants to predict the insurance charges based on the given Inputs example age, sex, etc.

Input-> age, bmi, children, sex, smoker
Output-> charges

Total number of rows and columns (1338 rows × 6 columns) before pre-processing.

	age	sex	bmi	children	smoker	charges
0	19	female	27.900	0	yes	16884.92400
1	18	male	33.770	1	no	1725.55230
2	28	male	33.000	3	no	4449.46200
3	33	male	22.705	0	no	21984.47061
4	32	male	28.880	0	no	3866.85520
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	10600.54830
1334	18	female	31.920	0	no	2205.98080
1335	18	female	36.850	0	no	1629.83350
1336	21	female	25.800	0	no	2007.94500
1337	61	female	29.070	0	yes	29141.36030

1338 rows × 6 columns

Total number of rows and columns (1338 rows × 6 columns) after pre-processing.

	age	bmi	children	charges	sex_male	smoker_yes
0	19	27.900	0	16884.92400	False	True
1	18	33.770	1	1725.55230	True	False
2	28	33.000	3	4449.46200	True	False
3	33	22.705	0	21984.47061	True	False
4	32	28.880	0	3866.85520	True	False
...	...	...	...	...	...	...
1333	50	30.970	3	10600.54830	True	False
1334	18	31.920	0	2205.98080	False	False
1335	18	36.850	0	1629.83350	False	False
1336	21	25.800	0	2007.94500	False	False
1337	61	29.070	0	29141.36030	False	True

1338 rows × 6 columns

3) Mention the pre-processing method if you're doing any

Converting string (Sex, Smoker) to number – nominal data using get_dummies function

4) R2 value

1. Multiple Liner Regression R2 Value =0.78947903

2. SUPPORT VECTOR MACHINE: -

S.NO	HYPER PARAMETER	LINEAR	POLY	RBF	SIGMOID
		R2 value	R2 value	R2 value	R2 value
1	C10	0.462468	0.038716	-0.032273	0.039307
2	C100	0.628879	0.617956	0.320031	0.52761
3	C500	0.763105	0.826368	0.664298	0.444606
4	C1000	0.764931	0.856648	0.810206	0.28747
5	C2000	0.744041	0.860557	0.854776	-0.59395
6	C3000	0.741423	0.859893	0.866339	-2.124419

The SVM Regression use R^2 Value (RBF and hyper parameter(C3000)) = 0.866339

3. Decision Tree Regressor: -

S.No	*criterion*	*max_features*	*splitter*	R2 Value
1	squared_error	sqrt	best	0.5134345
2	squared_error	sqrt	random	0.696855
3	squared_error	Log2	best	0.738349
4	squared_error	Log2	random	0.652272
5	friedman_mse	sqrt	best	0.662422
6	friedman_mse	sqrt	random	0.72539
7	friedman_mse	Log2	best	0.713697
8	friedman_mse	Log2	random	0.632829
9	absolute_error	sqrt	best	0.605655
10	absolute_error	sqrt	random	0.711887
11	absolute_error	Log2	best	0.758961
12	absolute_error	Log2	random	0.720418
13	poisson	sqrt	best	0.687067
14	poisson	sqrt	random	0.635849
15	poisson	Log2	best	0.721222
16	poisson	Log2	random	0.703589

The Decision Tree Regression use R^2 Value (absolute_error, Log2, best) =0.758961

4. Random Forest

S_No	criterion	max_features	n_estimators	R2 Value
1	squared_error	sqrt	10	0.851936
2	squared_error	sqrt	100	0.871984
3	squared_error	Log2	10	0.852507
4	squared_error	Log2	100	0.866981
5	friedman_mse	sqrt	10	0.863078
6	friedman_mse	sqrt	100	0.873748
7	friedman_mse	Log2	10	0.865468
8	friedman_mse	Log2	100	0.874149
9	absolute_error	sqrt	10	0.850794
10	absolute_error	sqrt	100	0.875751
11	absolute_error	Log2	10	0.859094
12	absolute_error	Log2	100	0.872328
13	poisson	sqrt	10	0.851107
14	poisson	sqrt	100	0.871667
15	poisson	Log2	10	0.849682
16	poisson	Log2	100	0.872961

Random Forest use R^2 Value (absolute_error, sqrt, n_estimators=100) = 0.875751

Final Model selected is Random Forest the output value which is nearest to 1 compare to other model.